

Development of Blood Oxygen Level, Heart Rate And Temperature Monitoring System by Using ESP32

Robiah Ahmad
Department of Engineering and Technology
Universiti Teknologi Malaysia
54100 Jalan Sultan Yahya Petra
Kuala Lumpur, Malaysia
robiahahmad@utm.my

Hazilah Mad Kaidi
Department of Engineering and Technology
Universiti Teknologi Malaysia
54100 Jalan Sultan Yahya Petra
Selangor, Malaysia
hazilah.kl@utm.my

Muhammad Noor Nordin
Medical Engineering Technology Section
Universiti Kuala Lumpur British Malaysian Institute
Selangor, Malaysia
muhammadnoor@unikl.edu.my

Aizat Faiz Ramli
Electronics Technology Section
Universiti Kuala Lumpur British Malaysian Institute
Selangor, Malaysia
aizatfaiz@unikl.edu.my

Mohd Azlan Abu
Malaysian Japan International Institute of Technology
Universiti Teknologi Malaysia
54100 Jalan Sultan Yahya Petra
Kuala Lumpur, Malaysia
mohdazlan.abu@utm.my

Yosof Kadase
Electronic Technology Section
Universiti Kuala Lumpur British Malaysian Institute
Selangor, Malaysia
yosof@unikl.edu.my

Abstract— Telehealth plays an even greater role in global healthcare delivery today. Ever since the pandemic strikes the world, the use of healthcare monitoring systems has a massive boost significantly. This paper aims to develop a prototype of a health monitoring device that can be put at a patient home to replace traditional physical checkups at the hospital by using MAX30102 PPG module and Infrared temperature sensor with the objective to derive PPG signal into basic health parameters such as heart rate and oxygen saturation from the patient through MAX30102 and to test the accuracy of those parameters with hospital patient monitor. The experiment is done by comparing each parameter for 30 seconds in both devices using index fingers coming from 13 participants. From the results obtained, although heart rate suffers motion artefact compared to the oxygen saturation, each accuracy is still excellent with 89.2 and 97.6 percent. In conclusion, MAX30102 PPG and ECG module can get initial health reading through the PPG signal and solve the problems mentioned.

Keywords—Telehealth, ESP32, monitoring

I. INTRODUCTION

The Internet of Things (IoT) is a network of connected devices and sensors that extends the internet and other network connections. For example, it can connect to alarms, gates, cars, and other objects with high-level determining and scientific capabilities. Many industries, including automotive [1], agriculture [2], [3], healthcare [4]–[6] and many more have embraced the IoT. The healthcare industry is one such trending field that provides good health services to all people worldwide.

Health by nature, is an illness or injury-free indication. Advancing biomedical technology and creating a longer life span is crucial to maintaining the quality of life. Healthcare is the advancement of health in diverse medical fields through prevention, diagnosis, and treatment offered by health practitioners. In the last 50 years, people aged 60 and above have tripled numbers and are predicted to be almost two billion by 2050 [7].

In the 21st century, chronic illnesses such as coronary conditions, heart failure (HF), high blood pressure, or obesity grew along with growth technologies in different capitalist

sectors such as the fast-food industry or unhealthy lifestyles. In particular, 80 per cent of the 35 million deaths associated with chronic diseases for 2019 would occur in regions with low and medium incomes. Men and women are infected younger than in countries with high incomes. In the next ten years, premature deaths are estimated to cost billions of dollars in countries such as China, India and the Russian Federation. In several nations, the urgency of risk mitigation and disease prevention strategies is working, and a series of low-cost, high-impact initiatives can be applied step-by-step [8].

Developed countries such as the United States, the United Kingdom, and Canada are experiencing a nursing shortage as the number of critically ill patients rises. The issue gets even more complicated. Increasing the number of chronically sick patients and the shortage of healthcare staff will increase the cost of services in the healthcare sector. A variable method is required to define patient management supervision and treatment while minimizing costs by delivering home health monitoring and care, called home telemonitoring.

One of the technologies that can assist in the health monitoring system is the IoT. Blood oxygen saturation, pulse rate, body temperature, heart rate, and body temperature are all vital indicators that should be examined on a regular basis [9].

This paper is organized as follows: Section II reviews the previous related work on healthcare IoT systems. Section III describes the methodology of this prototype development and Section V explains the experiment that has been conducted and a brief discussion of the results. Section IV finally concludes.

II. RELATED WORKS

The human body consists of a signal which carries specific information about the physiological system and can be examined by measuring the vital signs. Based on the analysis of several previous research, the trend is much more focused on measuring blood oxygen (SpO₂) [10], heart rate and temperature measurement. While other researchers also include the measurement of blood pressure and ECG measurement [11].

In the field of medical science, some significant work has been done using IoT to monitor patient health. Examples of works related to this field are explained below.

In an IoT network, Tamilselvi et al., 2020 [12] built a health monitoring system that can track a patient's fundamental symptoms such as heart rate, oxygen saturation percentage, body temperature, and eye movement. Heartbeat, SpO₂, Temperature, and Eye Blink sensors were employed as capturing elements, while an Arduino-UNO was used as a processing device. Although the developed system was deployed, there are yet defined performance measures for any of the patients.

In the IoT setting, Acharya et al. 2020 [13] presented a healthcare monitoring kit. Heartbeat, ECG, body temperature, and respiration were among the basic health metrics tracked by the designed system. Pulse sensor, temperature sensor, blood pressure sensor, ECG sensor, and Raspberry Pi are the main hardware components employed here. Sensor data was captured and forwarded to a Raspberry Pi for processing before being sent back to the IoT network. The system's biggest flaw is that no data visualisation interfaces have been created.

Most of the suggested health monitoring systems operate offline [14]. The online system is preferred to boost system efficiency. Some suggested online systems use radio frequency (RF) [15] to monitor a patient's condition at a remote node, although this does not cover a large region.

The data from the sensors can be collected using a variety of methods. The raw data from the sensors usually require an interface to make it readable. The detection of all four sensors has been interfaced with Arduino in various works [16], [17].

The Arduino generally features a Wi-Fi connection through an external module, making data transmission to web services easier. Other articles, on the other hand, have used an ESP32 as an interface to read the data. In addition, the ESP32 microcontroller features built-in Wi-Fi and Bluetooth dual-mode [18]. Aside from that, the ESP32 has a larger memory capacity than the Arduino [19].

Some proposed systems do not have a warning system when the readings are out of the ordinary. Some of them lack a built-in application that can show real-time readings. These functionalities are included in our suggested system, and the various sensors are examined in the study to find the best performance.

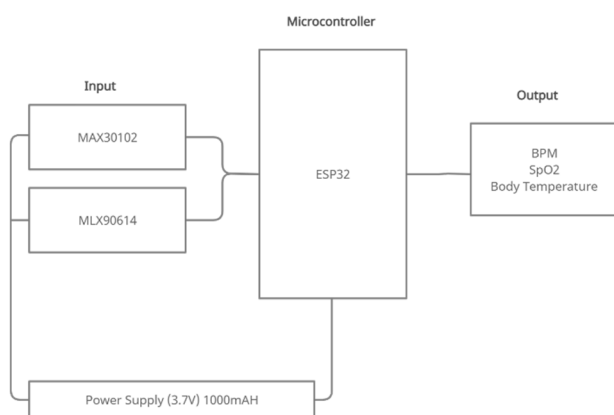


Fig. 1. A block diagram of the proposed system

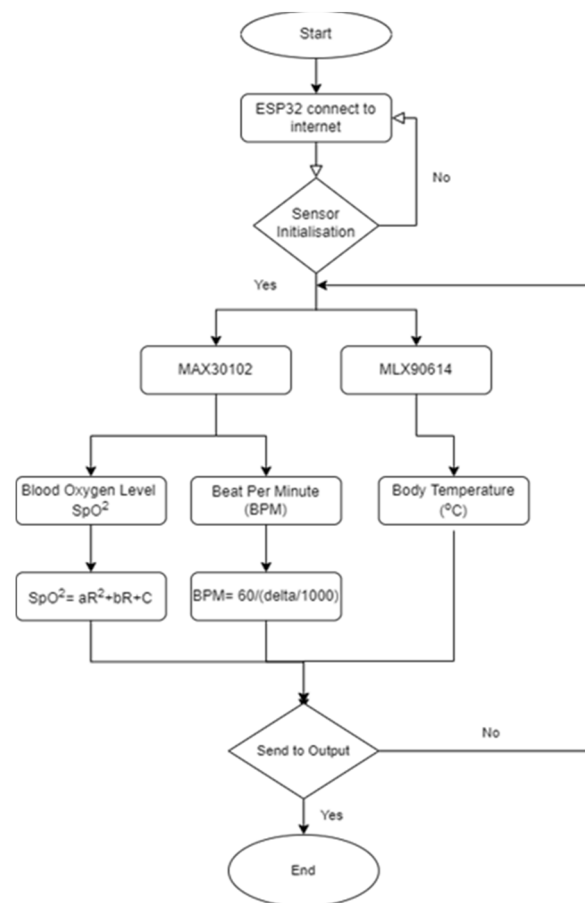


Fig. 2. A flow chart of the proposed system

III. PROPOSED SYSTEM

A health monitoring system using photoplethysmography (PPG) and Infrared (IR) sensors is proposed for this project implementation to acquire patient health information and transfer its data to health care providers. This system utilizes ESP32 as the system processor, Maxim MAX30102 PPG module as in [20] and MLX90614 IR Temperature module. The output is to be used as numerical details and can be used by users when necessary in monitoring health conditions. The patient's device creation would position the index finger at the top of the photoplethysmography and forehead for the IR sensor. The sensor will then use infrared and red monitoring to detect the user's heart rate, blood oxygen level, and temperature. The data collected will be displayed in android apps for patients or health experts.

Based on Fig. 1, the health monitoring system would consist of two inputs which are a PPG and an IR temperature sensor. The PPG is used for heart rate and blood oxygen saturation. Meanwhile, the second input is used to obtain the subject's body temperature. The power supply for this project will be from a LIPO rechargeable battery (1100mAH).

After these data have been collected, the sensor input is in raw signal data. Then, specific coding would be used in ESP32 using *tensorflow* microprocessor with low noise receive amplifier, power amplifier, filters, and power-management modules to translate these signals into numerical numbers. Then send those data outputs which show reading of BPM, SpO₂ and °C.

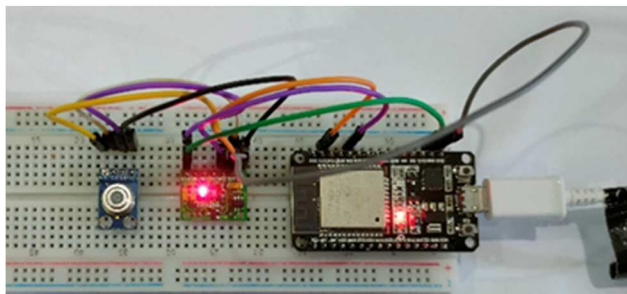


Fig. 3. The prototype of the proposed system

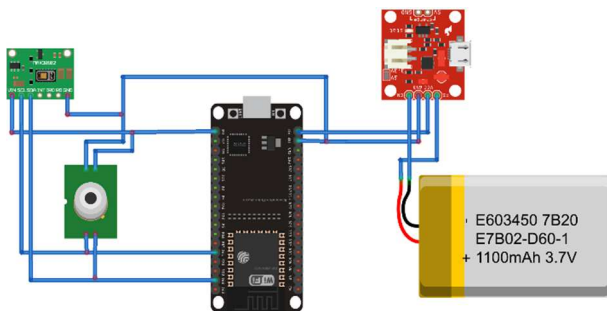


Fig. 4 Circuit diagram for the proposed system

The project flow (Fig. 2) will start when ESP32 is connected to the internet and cloud systems application. Then, MAX30102 and MLX90614 temperature sensors will start to initialize (Fig. 3). If those sensors are unsuccessful to initialize, then the system will reboot. Once both of the sensors have initialized, the system will check if the required condition is met by identifying whether user index finger is on the sensor. In addition, if the condition is met, both sensors will proceed to send the information to output. If no output is shown or zero reading is captured, the system will loop back to retain the sensor information.

Fig. 4. show a complete circuit diagram for the actual system. The sensor including MAX30102 and MLX90614 will be power by ESP32 3.3V pin. ESP32 will draw the power from LIPO rechargeable battery through the battery charger module. The output system can be viewed directly to Arduino IDE serial monitor or through dedicated cloud systems.

A. Measurement Algorithm

Heart rate from MAX30102 is a derivation from the PPG signal. It is based on the systolic and diastolic peaks as indicated in Fig. 5. The measurement is done by counting the systolic peak for each cardiac cycle for one second and then multiplying those peaks by 60 seconds by using the formula $60 \times (\text{delta}/1000)$.

MAX30102 provides PPG that contains infrared and red led to detect blood volume changes in the microvascular bed of tissue. This kind of detection has been used in SPO2 devices in clinical care. In theory, MAX30102 can detect oxygen saturation in the human body by using infrared and red led to measure changes in the light absorption in the human body. This subsection explains how to read oxygen saturation in the human body using MAX30102.

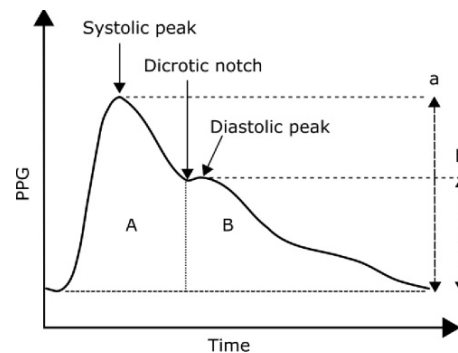


Fig. 5. Typical PPG waveform including systolic and diastolic peak

The key to creating oxygen saturation in the PPG signal lies in the infrared and red led ratio. The oxyhemoglobin contains oxygen when attracted to the 950nm wavelength of infrared. While non-oxyhemoglobin will be attracted to the 650nm wavelength of red light. From the ratio of both wavelength absorbance, oxygen saturation in human body can be calculated by using an equation

$$aR^2 + bR + c. \quad (1)$$

The a , b and c values are constant, which are $a=-16.67$, $b=8.33$ and $c=100$. So for R -value is coming from $R=\text{Red}/\text{Infrared}$. Value a , b and c are obtained through series of comparative measurement with medical grade SpO_2 sensor with reading obtain by our PPG sensor. Thus, measurement of heart rate is conducted by counting systolic peak for each cardiac cycle for ten second and then multiply those peaks by 6 to get beat per minute (BPM) measurement.

B. Accuracy Measurement

In order to ensure sensor accuracy while collecting user health information, a series of comparative measurements is conducted to ensure all information recorded is on par with certified available healthcare products in the market. To compare the reading of heart rate and blood oxygen level the selected product that been chosen is Philips patient monitor IntelliVue X3 (Fig. 6). Meanwhile, for body temperature, the selected device was a non-contact Infra-red thermometer by Duratool (DR3055) as shown in Fig. 7. The selected devices are certified by SIRIM and Malaysia Medical Device Authority (MDA).



Fig. 6. Philips patient monitor model IntelliVue X3



Fig. 7. Duratool DR3055 non-contact IR thermometer

IV. EXPERIMENT AND RESULT

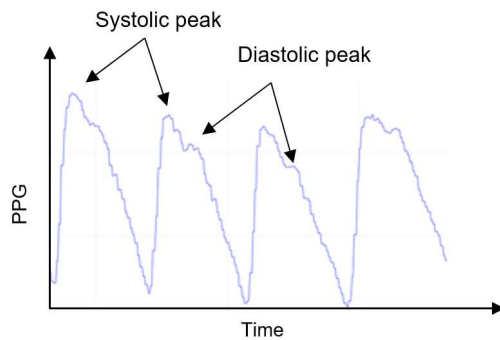


Fig. 8. MAX30102 raw PPG signal

The raw PPG signal produced by MAX30102 is shown in Fig. 8. However the signal is noisy due to increasing optical density and path length of illuminated tissue while decreasing the light intensity of the photodetector so that the PPG signal is correlated with blood volume. Therefore, it is hard to see systolic and diastolic peaks from the produced signal. To solve this issue, we create an analogue interface circuit as the op-amp will work as a current to voltage converter. Also, series of filter has been developed to remove excess of noisy PPG signal. For SpO_2 , the measurement conducted base on Eq (1). Meanwhile for the MLX90614 IR sensor, no changes are required to be done on raw signal and measurement was directly converted by microprocessor based on input voltage from the sensor.

A. Heart Rate

Table I compares the heart rate parameter between the patient monitor and the product prototype. There are 13 subjects from ages 22 to 24 years old, two subjects with age 24 years old, seven patients with age 22 years old and four patients with age 23 years old for all obtained data. The patient monitor's highest value is 105 BPM collected from subject 2, and the lowest is 63 BPM collected from subject 4. For the product prototype, the highest value is 101.5 BPM from subject 11, and the lowest is 27.5 BPM from subject 12.

The low reading in patient 12 is suspected due to poor sensor placement during heart rate measurement. All test subject has been instructed not to move their finger and in rest position throughout the measurement. The measurement was conducted for 2 minutes straight to ensure reading accuracy.

TABLE I. THE HEART RATE DATA COLLECTION

Subject	Age	Patient Monitor (%)	Heart Rate Prototype (%)	Error (%)
Subject 1	24	95	86	9
Subject 2	24	105	92	12
Subject 3	22	85	79	7
Subject 4	22	63	63.75	1
Subject 5	22	78	77.5	1
Subject 6	23	83	73.5	11
Subject 7	22	90	87.75	2.5
Subject 8	23	65	62.75	3
Subject 9	22	87	83	4.5
Subject 10	23	84	87.5	4
Subject 11	22	102	101.5	0.4
Subject 12	23	90	27.5	70
Subject 13	22	98	82	16

B. SpO_2

Table II tabulates the SpO_2 data collected from subjects. The patient monitor's highest value is 100% oxygen saturation collected from all subjects except for subjects 4 and 5. The lowest is 97% oxygen saturation collected from subject 5. The spo_2 data collection.

Subject	Age	Patient Monitor (%)	Product Prototype (%)	Error (%)
Subject 1	24	100	98	2
Subject 2	24	100	96	4
Subject 3	22	100	98	2
Subject 4	22	99	98	1
Subject 5	22	97	100	3
Subject 6	23	100	97.8	2.2
Subject 7	22	100	98	2
Subject 8	23	100	98.9	1
Subject 9	22	100	98.73	1.3
Subject 10	23	100	98.81	1.1
Subject 11	22	100	98	2
Subject 12	23	100	98.9	1
Subject 13	22	100	93	7

For the product prototype, the highest value is 100% oxygen saturation from subject 5, and the lowest is 93% from subject 13.

Subject 5 oxygen saturation on the product prototype is excellent. Unfortunately, the value on the patient monitor is 97% which is peculiar because the patient monitor is standardize medical equipment with better accuracy and filter than the product prototype. Upon investigation, subject 5's finger was moving inside the patient monitor finger clip, creating motion artefacts that increased light level changes. The inadequate signal is caused by motion artefact makes the patient monitor difficult to calculate oxygen saturation, as shown in Fig. 4.

C. Body Temperature

There are plenty of places to check the body's temperature, such as the forehead, mouth, and ear. However, this experiment will use the index finger to capture body temperature since the index finger has the most arteriovenous anastomoses or AVAs. It regulates vasoconstriction and vasodilatation more than the rest of the body surface. The finger also correlates with corneal temperature as blood flow is parallel with the finger and the eye.

Table III depicts the body temperature data for 13 subjects. The infrared thermometer's highest value is 36.7 °C collected from the subject, and the lowest is 33.5 °C from subject 10. For the product prototype, the highest value is 35.6 °C from subject 3, and the lowest is 28.71°C from subject 13.

TABLE II. THE BODY TEMPERATURE DATA COLLECTION

Subject	Age	Infrared thermometer (°C)	Product Prototype (°C)	Error (%)
Subject 1	24	35.7	34.5	3.3
Subject 2	24	35	34.7	0.8
Subject 3	22	36	35.6	1
Subject 4	22	35.8	34.8	2.7
Subject 5	22	36.7	35.3	3.8
Subject 6	23	34.3	34.71	1
Subject 7	22	34.7	33.4	4.8
Subject 8	23	35.6	35.31	0.8
Subject 9	22	33.7	32.6	3
Subject 10	23	33.5	32.45	3
Subject 11	22	34.1	33.4	2
Subject 12	23	34.3	28.71	16
Subject 13	22	34	30	11

The difference between the highest and lowest value of the infrared thermometer is 3.2 °C, and for the product prototype is 6.89 °C. The infrared thermometer average value is 34.83 °C and 33.33 °C for the product prototype. Meanwhile, the average error value is 5.2% with 95% accuracy.

As the discussion for results tabulated in Table I and II, the comparative measurement was conducted has shown a high percentage similarity with medical grade equipment. It calculated that for heart rate, the average similarity recorded at 89%, while for SpO₂ at 98%. The reason for much lower heart rate measurement accuracy, although coming from the same sensor MAX30102 is movement artefact sensitivity. The heart rate measurement is very sensitive to the motion of the test subject's finger. Another issue related to measurement is the method of both sensors in capturing the light produced by a red and infrared LED. For MAX30102, the sensor uses reflective method for detection of light while the patient monitor uses transmission of light through the finger [21].

Meanwhile, body temperature measurements indicate below range of normal body temperature around 36.5 to 37.5 °C. Both IR temperature sensors under performance mainly due to two major factor: distance of the subject with the sensor and the environment where measurement is taken. These two factors contribute massively on producing accurate body temperature measurements. Furthermore, it is reported by Hsuan *et al* [22] that measurement of core body temperature by using IR sensor probably requires strict standard operating procedure (SOP) due to this issue. They also suggest that all measurement related to IR sensor temperature has to be undergo the black box method to ensure accuracy.

V. CONCLUSION

In this system, user health monitoring can be achieved by collecting the physiological information from the various sensor, microcontroller ESP32 and output. The MAX30102 module has shown excellence in measuring heart rate and blood oxygen level parameters, scoring accuracy around 89% and 98%, respectively. While for IR temperature, the comparative studies show MLX90614 accuracy recorded at 95%. The system can accurately collect user health information as a guide for better health assessment and lifestyle. This system can be further improved with additional health information detection such as blood pressure, which is a more complex algorithm. In the near future, this particular system will change the healthcare monitoring system worldwide.

ACKNOWLEDGEMENT

The authors would like to express their gratitude to Universiti Teknologi Malaysia and Universiti Kuala Lumpur, which have provided research funding through Collaborative Research Grant Vot R.K130000.7356.4B416 and CRGS/2019/UniKL/03/1.

REFERENCES

- [1] D. Pejic, V. Krizanovic, and K. Grgic, "Detection of attacks and intrusions on automotive engine IoT sensors," in *Proceedings of the 16th International Conference on Telecommunications, ConTEL 2021*, Jun. 2021, pp. 134–141. doi: 10.23919/ConTEL52528.2021.9495981.
- [2] H. M. Kaidi *et al.*, "Internet of Things: A Monitoring and Control System for Rockmelon Farming," *International Journal of Integrated Engineering*, vol. 12, no. 6, pp. 55–61, 2020, doi: 10.30880/ijie.2020.12.06.007.
- [3] A. N. Harun, R. Ahmad, N. Mohamed, A. R. A. Rahim, and H. M. Kaidi, "Morphological and physiological responses of brassica chinensis on different far-red (FR) light treatments using internet-of-things (IoT) technology," *Agriculture (Switzerland)*, vol. 11, no. 8, Aug. 2021, doi: 10.3390/agriculture11080728.
- [4] R. Hegde, S. Ranjana, and C. D. Divya, "Survey on Development of Smart Healthcare Monitoring System in IoT Environment," in *Proceedings - 5th International Conference on Computing Methodologies and Communication, ICCMC 2021*, Apr. 2021, pp. 395–399. doi: 10.1109/ICCMC51019.2021.9418405.
- [5] K. Anjali, R. Anand, S. D. Prabhu, and R. S. Geethu, "IoT based Smart Healthcare System to Detect and Alert Covid Symptom," in *Proceedings of the 6th International Conference on Communication and Electronics Systems, ICCES 2021*, Jul. 2021, pp. 685–692. doi: 10.1109/ICCES51350.2021.9488952.
- [6] N. Zholdas, O. Postolache, and M. Mansurova, "Health Monitoring System Using Internet of Things," Apr. 2021. doi: 10.1109/SIST50301.2021.9465928.
- [7] G. Paré, M. Jaana, and C. Sicotte, "Systematic Review of Home Telemonitoring for Chronic Diseases: The Evidence Base," *Journal of the American Medical Informatics Association*, vol. 14, no. 3, pp. 269–277, May 2007. doi: 10.1197/jamia.M2270.
- [8] World Health Organization. and Public Health Agency of Canada., *Preventing chronic diseases : a vital investment*. World Health Organization, 2005.
- [9] S. Nookhao, V. Thananant, and T. Khunkhao, "Development of IoT Heartbeat and Body Temperature Monitoring System for Community Health Volunteer," in *DAMT & NCON 2020 : International Conference on Digital Arts, Media and Technology (DAMT) and ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunications Engineering (NCON)*, 2020, pp. 106–109.
- [10] C. K. Tseng *et al.*, "Application of a minimized wearable device combined with SpO2 and ECG sensors to detect stenosis or occlusion of arteriovenous fistula/graft, progression of arteriosclerosis and arrhythmia," in *Proceedings - 2018 7th International Symposium on Next-Generation Electronics, ISNE 2018*, Jun. 2018, pp. 1–4. doi: 10.1109/ISNE.2018.8394718.
- [11] A. N. N. Chamim, J. Rinaldi, Y. Ardiyanto, I. Iswanto, and A. Ma'Arif, "Heart Rate and Body Temperature Monitoring Based on Android Operating System," in *Proceeding - 2020 2nd International Conference on Industrial Electrical and Electronics, ICIEE 2020*, Oct. 2020, pp. 143–148. doi: 10.1109/ICIEE49813.2020.9276750.
- [12] V. Tamilselvi, S. Sribalaji, P. Vigneshwaran, P. Vinu, and J. Geetha Ramani, "IoT Based Health Monitoring System" in *2020 6th International Conference on Advanced Computing & Communication Systems (ICACCS)*, 2020, pp. 386–389.
- [13] A. D. Acharya and S. N. Patil, "IoT based Health Care Monitoring Kit," in *Proceedings of the 4th International Conference on Computing Methodologies and Communication, ICCMC 2020*, Mar. 2020, pp. 363–368. doi: 10.1109/ICCMC48092.2020.ICCMC-00068.
- [14] H. F. Nweke, Y. W. Teh, G. Mujtaba, and M. A. Al-garadi, "Data fusion and multiple classifier systems for human activity detection and health monitoring: Review and open research directions," *Information Fusion*, vol. 46, pp. 147–170, Mar. 2019, doi: 10.1016/j.inffus.2018.06.002.
- [15] M. F. Ahmed, M. O. Ali, M. H. Rahman, and Y. M. Jang, "Real-time health monitoring system design based on optical camera communication," in *International Conference on Information Networking*, Jan. 2021, vol. 2021-January, pp. 870–873. doi: 10.1109/ICOIN50884.2021.9334018.
- [16] R. Sameh, M. Genedy, A. Abdeldayem, and M. H. Abdel azeem, "Design and Implementation of an SPO2 Based Sensor for Heart Monitoring Using an Android Application," *Journal of Physics: Conference Series*, vol. 1447, no. 1, p. 012004, Jan. 2020, doi: 10.1088/1742-6596/1447/1/012004.
- [17] Md. S. Mahmud, H. Wang, A. M. Esfar-E-Alam, and H. Fang, "A Wireless Health Monitoring System Using Mobile Phone Accessories," *IEEE Internet of Things Journal*, vol. 4, no. 6, pp. 2009–2018, Dec. 2017, doi: 10.1109/JIOT.2016.2645125.
- [18] M. Islam, A. Rahaman, and R. Islam, "Development of Smart Healthcare Monitoring System in IoT Environment," vol. c, 2020.
- [19] S. K., P. A. S., P. U., P. S., and S. P. V., "Patient health monitoring system using IoT," *Materials Today: Proceedings*, Jun. 2021, doi: 10.1016/j.matpr.2021.06.188.
- [20] K. Rajguru, P. Tarpe, V. Aswar, K. Bawane, S. Sorte and R. Agrawal, "Design and Implementation of IoT based sleep monitoring system for Insomniac people," 2022 Second International Conference on Artificial Intelligence and Smart Energy (ICAIS), 2022, pp. 1215–1221, doi: 10.1109/ICAIS53314.2022.9742803.
- [21] F. Guede-Fernández, V. Ferrer-Mileo, M. Mateu-Mateus, J. Ramos Castro, and M. Fernández-Chimeno, "A photoplethysmography smartphone-based method for heart rate variability assessment: Device model and breathing influences," *Biomedical Signal Processing and Control*, vol. 57, p. 101717, 2020.
- [22] Chen, H., Chen, A. and Chen, C., 2020. Investigation of the Impact of Infrared Sensors on Core Body Temperature Monitoring by Comparing Measurement Sites. *Sensors*, 20(10), p.2885.